

Experiment:5

Construct a scheduling program with C that selects the waiting process with the highest priority to execute next.

Code:-

```
#include <stdio.h>

struct priority_scheduling
{
    char process_name;
    int burst_time;
    int waiting_time;
    int turn_around_time;
    int priority;
};

int main()
{
    int number_of_process;
    int total = 0;
    int ASCII_number = 65;
    int position;

    float average_waiting_time, average_turnaround_time;

    printf("Enter the total number of Processes: ");

    scanf("%d", &number_of_process);

    struct priority_scheduling process[number_of_process], temp_process;

    printf("\nPlease Enter the Burst Time and Priority of each process:\n");

    for (int i = 0; i < number_of_process; i++)
    {
```

```
    process[i].process_name = (char)ASCII_number;
printf("\nEnter the details of Process %c\n", process[i].process_name);
printf("Enter Burst Time: ");
    scanf("%d", &process[i].burst_time);

    printf("Enter Priority: ");
    scanf("%d", &process[i].priority);

    ASCII_number++;
}
for (int i = 0; i < number_of_process; i++)
{
    position = i;
    for (int j = i + 1; j < number_of_process; j++)
    {
        if (process[j].priority > process[position].priority)
            position = j;
    }
    temp_process = process[i];
    process[i] = process[position];
    process[position] = temp_process;
}
process[0].waiting_time = 0;

for (int i = 1; i < number_of_process; i++)
{
    process[i].waiting_time = 0;
```

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        for (int j = 0; j < i; j++)
            process[i].waiting_time += process[j].burst_time;
total += process[i].waiting_time;
    }
average_waiting_time = (float)total / number_of_process;
total = 0;
printf("\nProcess\tBT\tPriority\tWT\tTAT\n");
for (int i = 0; i < number_of_process; i++)
{
    process[i].turn_around_time =
        process[i].burst_time + process[i].waiting_time;
total += process[i].turn_around_time;
printf("%c\t%d\t%d\t\t%d\t%d\n",
        process[i].process_name,
        process[i].burst_time,
        process[i].priority,
        process[i].waiting_time,
        process[i].turn_around_time);
}
average_turnaround_time = (float)total / number_of_process;

printf("\nAverage Waiting Time = %.2f", average_waiting_time);
printf("\nAverage Turnaround Time = %.2f\n", average_turnaround_time);

return 0;
}

```

Output:

```
Enter the total number of Processes: 3

Please Enter the Burst Time and Priority of each process:

Enter the details of Process A
Enter Burst Time: 7
Enter Priority: 5

Enter the details of Process B
Enter Burst Time: 3
Enter Priority: 8

Enter the details of Process C
Enter Burst Time: 3
Enter Priority: 2

Process BT      Priority      WT      TAT
B          3          8          0          3
A          7          5          3         10
C          3          2         10         13

Average Waiting Time = 4.33
Average Turnaround Time = 8.67

...Program finished with exit code 0
Press ENTER to exit console.
```